

Software Requirements Specification

AI-Based Deepfake Interaction System for Entertainment

Course / Context	DSP Project - SRS Document
Project Topic	Deepfake Application
System Title	AI-Based Deepfake Interaction System for Entertainment

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1 Introduction

1.1 Purpose of the System

The purpose of this software system is to design an AI-based deepfake interaction platform for entertainment and academic demonstration. The system enables users to generate celebrity-style synthetic speech, perform face swapping in video, simulate personality-based conversations, and run interactive meeting-like scenarios while preserving visible disclosure that the content is AI generated.

1.2 Scope of the System

The proposed system is intended for controlled entertainment, parody, and educational experimentation. It covers text-to-speech generation using predefined voice models, AI video face swap using selected public-figure templates, real-time or near real-time conversation simulation, downloadable outputs, and evaluation of generated media using objective quality metrics. The system is not intended for identity fraud, impersonation without disclosure, or unauthorized public misuse.

2 System Overview

The system operates as an interactive multimedia platform in which the user selects a predefined celebrity-style profile, enters text or live audio/video input, configures generation options, previews the result, and exports the final media with required safeguards.

The major components of the system are the Voice Cloning Module, AI Video Face Swap Module, AI Conversation Module, Meeting Simulation Module, Quantitative Evaluation Module, Watermark and Safety Control Layer, and User Interface Module.

The main objectives of the system are to provide entertaining and controllable deepfake interaction, demonstrate DSP and AI concepts in practice, measure output quality using academic metrics, and ensure that generated media is clearly marked as synthetic.

The operating system for the proposed software shall be Windows 10/11 or Ubuntu Linux running on a standard personal computer. Optional webcam and microphone support may be used for live interaction features.

Major System Modules

Module	Role in the System
Voice Cloning Module	Generates celebrity-style speech from text and optionally converts live voice input.
AI Video Face Swap Module	Detects, aligns, and swaps faces while preserving expression, pose, and lighting.
AI Conversation Module	Simulates text and voice dialogue in the selected public-figure style.
Meeting Simulation Module	Places the generated character in a virtual meeting or scenario-based interaction.
Quantitative Evaluation Module	Computes MCD, SNR, SSIM, PSNR, and latency for academic assessment.
Watermark and Safety Control Layer	Applies visual and audio watermarking and enforces parody/AI-generated disclosure.
User Interface Module	Provides a simple and understandable interface for selecting models, entering prompts, previewing results, and exporting outputs.

3 Functional Requirements

3.1 Voice Cloning Module

- 3.1.1** The system shall provide a list of predefined celebrity voice models.
- 3.1.2** The system shall convert user-provided text into synthesized speech using the selected voice model.
- 3.1.3** The system shall allow the user to adjust speaking rate, pitch, and emotional tone.
- 3.1.4** The system may accept real-time microphone input for live voice conversion.
- 3.1.5** The system shall generate downloadable audio output in WAV or MP3 format.
- 3.1.6** The system shall embed an inaudible watermark in generated audio output.

3.2 AI Video Face Swap Module

- 3.2.1** The system shall provide predefined face models of selected public figures.
- 3.2.2** The system shall detect and align faces in uploaded video or webcam input before swapping.
- 3.2.3** The system shall replace the source face with the selected target face while preserving facial expression, head pose, and lighting consistency.
- 3.2.4** The system shall support real-time or near real-time deepfake rendering for meeting simulation mode.

3.2.5 The system shall display a permanent on-screen visual watermark indicating synthetic generation during video output.

3.3 AI Conversation Module

3.3.1 The system shall simulate text-based conversation in the style of the selected public figure.

3.3.2 The system shall support live two-way dialogue in both text chat and voice-based interaction modes.

3.3.3 The system shall log generated conversations for academic evaluation and system testing.

3.3.4 The system shall store conversation logs securely and allow authorized retrieval by the system administrator.

3.4 Meeting Simulation Module

3.4.1 The system shall provide a virtual meeting room interface in which the AI-generated character appears as a participant.

3.4.2 The system shall allow the user to choose predefined scenarios including comedy interview, job interview parody, friendly debate, and motivational speech.

3.4.3 The system may optionally support multi-character mode for more than one AI-generated participant in the same session.

3.5 Quantitative Evaluation Module

3.5.1 The system shall compute Mel-Cepstral Distortion (MCD) for generated voice output.

3.5.2 The system shall compute Signal-to-Noise Ratio (SNR) for generated audio quality analysis.

3.5.3 The system shall compute Structural Similarity Index (SSIM) for video quality evaluation.

3.5.4 The system shall compute Peak Signal-to-Noise Ratio (PSNR) for reconstructed video quality evaluation.

3.5.5 The system shall measure end-to-end system latency for real-time simulation mode.

3.5.6 The system shall present evaluation results in a structured and readable report interface.

3.6 User Interface Module

3.6.1 The system shall provide a celebrity selection panel for voice model, face model, and conversation style.

3.6.2 The system shall display a real-time preview before the final output is exported.

- 3.6.3** The system shall allow the user to download the generated result after processing is completed.
- 3.6.4** The system shall restrict direct public auto-sharing integrations in order to reduce uncontrolled distribution.
- 3.6.5** The system shall validate input media before processing and report errors clearly when files are unsupported or corrupted.

4 Non-Functional Requirements

4.1 Performance Requirements

- 4.1.1** The system shall provide preview responses within 2 seconds for user interaction under normal hardware conditions.
- 4.1.2** The system shall support real-time processing with less than 500ms latency for live meeting simulation when optimized models are enabled.
- 4.1.3** The system shall complete offline generation tasks using no more than 4GB of RAM on a standard personal computer.

4.2 Reliability Requirements

- 4.2.1** The system shall maintain stable processing for continuous sessions of up to 2 hours without crashing.
- 4.2.2** The system shall preserve generated files and logs in the event of non-critical module failure whenever recovery is possible.
- 4.2.3** The system shall gracefully recover from 100% of invalid media input attempts without terminating the application.

4.3 Usability Requirements

- 4.3.1** The system shall allow users to complete a standard deepfake generation configuration within 3 menu clicks.
- 4.3.2** The system's evaluation metrics shall be presented with integrated informational tooltips explaining MCD, SNR, SSIM, and PSNR for student comprehension.

4.4 Security and Privacy Requirements

- 4.4.1** The system shall restrict administrative settings and log access to authorized users only.
- 4.4.2** The system shall store temporary input media and logs using secure local storage practices.
- 4.4.3** The system shall avoid exposing raw training assets or internal model files to end users.

5 System Constraints

5.1 Budget Constraints

- 5.1.1** The system shall be developed primarily using open-source tools, libraries, and frameworks in order to minimize development cost.
- 5.1.2** The project shall avoid the use of commercial or paid software licenses whenever possible.

5.2 Hardware Constraints

- 5.2.1** The system shall operate on standard personal computers without requiring specialized hardware.
- 5.2.2** Optional webcam, microphone, and mid-range GPU acceleration may be used to improve live interaction performance, but the baseline design shall remain executable on common laboratory or student PCs.

5.3 Technology Constraints

- 5.3.1** The system shall enforce watermarking and disclosure mechanisms that cannot be removed or bypassed.

5.4 Ethical and Operational Constraints

- 5.4.1** The system shall not be designed for deception, identity theft, or undisclosed impersonation.
- 5.4.2** The system shall operate under a mandatory disclosure policy, preventing the generation of any media without clear synthetic attribution.
- 5.4.3** The system shall use predefined public-figure models and publicly available datasets only within the limits of academic demonstration and lawful use.